

Exercise 11: Nonlinear Programming (Lagrange + KKT + SVM)

Optimization in Machine Learning

Reproducibility

Produced with Python 3.10 and R 4.4.1. To reproduce, install the pinned package versions below.

Python

```
pip install numpy==2.0.2 matplotlib==3.10.8 cvxpy==1.7.5
```

R

```
install.packages("remotes")
remotes::install_version("CVXR", "1.8.2")
remotes::install_version("ggplot2", "3.5.1")
remotes::install_version("mlr3", "1.6.0")
```

Problem 1: Lagrange multipliers

Solve the constrained optimization problem

$$\min_{(x,y) \in \mathbb{R}^2} x + 2y \quad \text{s.t.} \quad x^2 + 4y^2 = 4$$

by Lagrange multipliers.

Problem 2: Nonlinear SVM

We have 200 binary observations from a *moons* distribution (two interleaved half-circles) — not linearly separable. The standard SVM trick: lift to a higher-dimensional feature space via $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^l$ and learn a linear separator there. The primal soft-margin SVM in feature space is

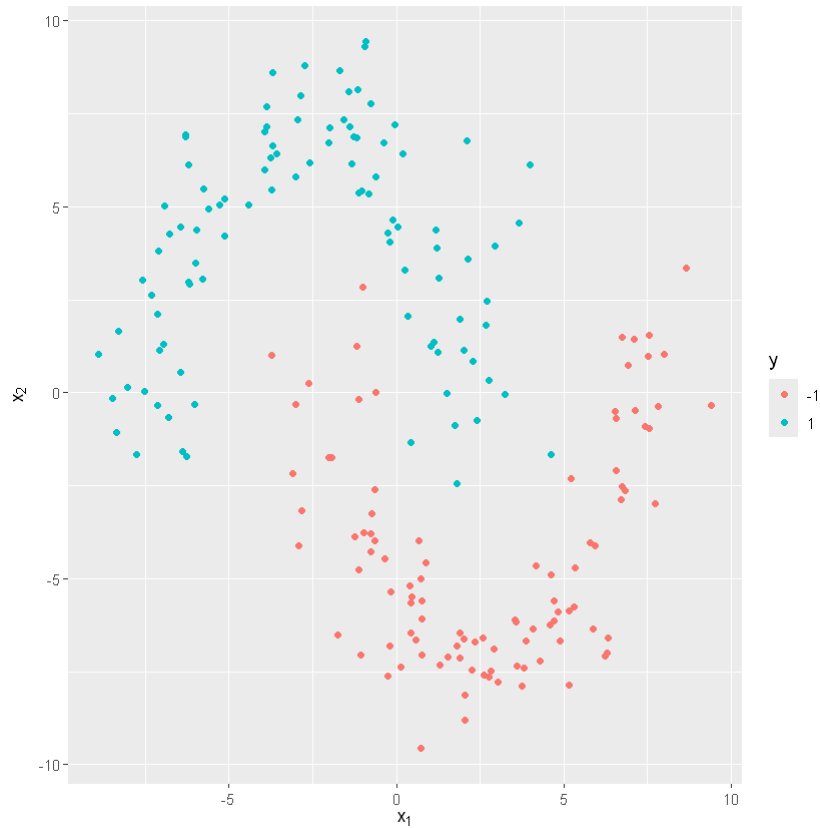
$$\min_{\theta, \theta_0, \zeta} \frac{1}{2} \|\theta\|^2 + C \sum_{i=1}^n \zeta^{(i)} \quad \text{s.t.} \quad y^{(i)} (\langle \theta, \phi(\mathbf{x}^{(i)}) \rangle + \theta_0) \geq 1 - \zeta^{(i)}, \quad \zeta^{(i)} \geq 0 \quad \forall i.$$

R

```
library(ggplot2)
library(CVXR)

# 200 nonlinearly separable observations from a moons distribution.
moon_data <- read.csv("data/moons.csv")
n <- nrow(moon_data)
moon_data$y_dec <- as.factor(moon_data$y)

ggplot(moon_data, aes(x = x1, y = x2, color = y_dec)) +
  geom_point(size = 1.5) +
  xlab(expression(x[1])) + ylab(expression(x[2])) +
  labs(color = expression(y))
```



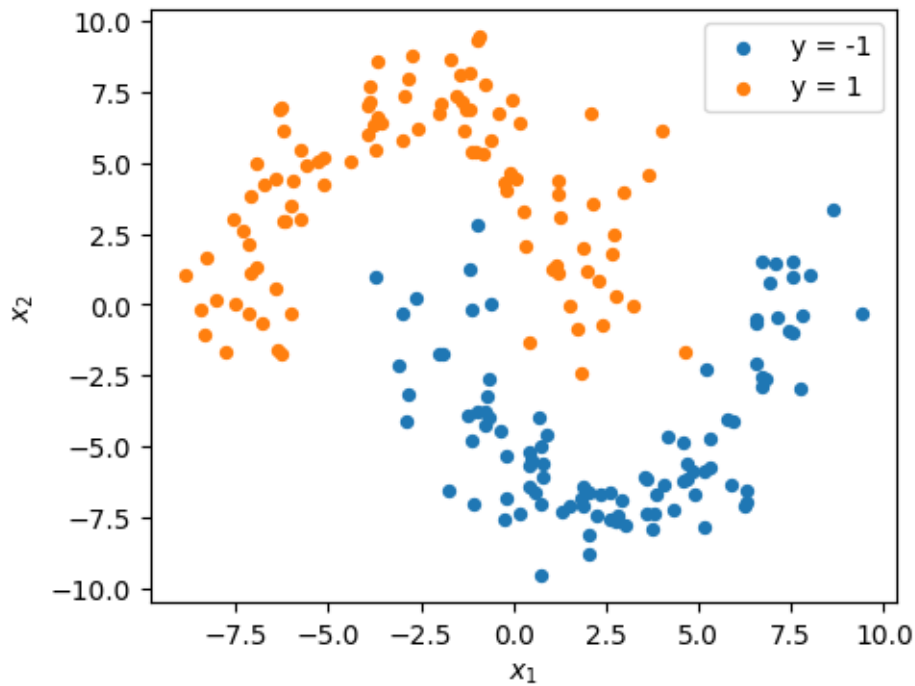
Python

```
import numpy as np
import matplotlib.pyplot as plt
import cvxpy as cp

# 200 nonlinearly separable observations from a moons distribution.
data = np.genfromtxt("data/moons.csv", delimiter=",", names=True)
X = np.column_stack([data["x1"], data["x2"]])
y = data["y"].astype(float)
n = X.shape[0]

fig, ax = plt.subplots(figsize=(5, 4))
for label in (-1, 1):
    mask = y == label
    ax.scatter(X[mask, 0], X[mask, 1], label=f"y = {label}", s=18)
ax.set_xlabel(r"$x_1$"); ax.set_ylabel(r"$x_2$")
```

```
ax.legend()
plt.show()
```



(a) Lagrangian

Write down the Lagrangian of the nonlinear SVM.

(b) Solve the primal SVM

Solve the primal nonlinear SVM with $C = 1$ and a third-order polynomial transformation (without intercept)

$$\phi(\mathbf{x}) = (x_1, x_2, x_1^2, x_2^2, x_1x_2, x_1^2x_2, x_1x_2^2, x_1^3, x_2^3)^\top \in \mathbb{R}^9$$

using CVXR (R) or cvxpy (Python).

(c) KKT conditions

State the KKT conditions of the general nonlinear SVM.

(d) Dual form

Derive the dual form of the nonlinear SVM. State an advantage of the dual over the primal.

Hint: use the KKT conditions to eliminate the primal variables from the Lagrangian.

(e) Solve the dual SVM

Solve the dual form from (b) via `CVXR/cvxpy`.

Hint 1: For a polynomial transformation ϕ of order l (without intercept) there exists an invertible diagonal $\mathbf{D} \in \mathbb{R}^{l \times l}$ such that $\langle \mathbf{D}\phi(\mathbf{x}), \mathbf{D}\phi(\mathbf{z}) \rangle = (\mathbf{x}^\top \mathbf{z} + 1)^l - 1$.

Hint 2: Add $10^{-7}\mathbf{I}$ to the kernel matrix to ensure invertibility / positive-definiteness.